Jonathan (Wooyoung) Chung

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Professional Summary

Ph.D. student in machine learning and control (first-author papers at ACC and CDC; co-first-author work under review at ICLR and ACC) working on unsupervised reinforcement learning, information-theoretic model reduction, and learning on Lie groups, with GPU-scale RL pipelines in JAX/PyTorch and MuJoCo-MJX. Seeking a Summer 2027 ML internship in robotics.

Education

Doctor of Philosophy in Computer Science — **Lubbock, Texas**
Texas Tech University, Edward E. Whitacre Jr. College of Engineering — *January 2025 – Present*
- **Advisor:** Prof. Stas Tiomkin. **Award:** DGSA Fellowship. **Coursework:** Control Theory II, Lie Groups, Algebraic Geometry

Master of Science in Artificial Intelligence — **San Jose, California**
San Jose State University, Charles W. Davidson College of Engineering — *January 2022 – May 2024*
- **Coursework:** Deep Learning, Machine Learning, Reinforcement Learning, Data Mining, Big Data Analysis

Bachelor of Science in Computer Science — **Davis, California**
University of California, Davis, College of Letters & Science — *September 2019 – August 2022*

Publications

- **W. Chung***, T. Shah*, V. Makarenko, J. Wachs, S. Tiomkin. "Bellman Meets Lyapunov: Unsupervised Reinforcement Learning via Mastering Chaos." *Under review at ICLR 2027* (*equal contribution). arXiv Website Code
- **W. Chung***, T. Shah*, V. Makarenko, S. Tiomkin. "A Unified Framework for Empowerment and Predictive Control." *Under review at ACC 2027* (*equal contribution). arXiv Code
- **W. Chung**, S. Tiomkin. "Dynamics Aware Past-Future Information Bottleneck." *IEEE Conf. on Decision and Control (CDC)*, 2026, to appear. Code
- **W. Chung**, D. Polani, S. Tiomkin. "Dimensionality Reduction of Dynamics on Lie Groups via Structure-Aware Canonical Correlation Analysis." *American Control Conference (ACC)*, 2024, pp. 439–446. DOI Code

Experience

Texas Tech University — **Lubbock, Texas**
Graduate Research Assistant — *January 2025 – Present*
- **Unsupervised RL from dynamical-systems theory** (ICLR 2027, under review; co-first author): co-developed Forward CIP, a forward-in-time decomposition of the sum of positive Lyapunov exponents into a per-step Markovian reward, making a dynamics-grounded intrinsic objective optimizable with standard RL (exactness proven via control–estimation duality). PPO agents trained in MuJoCo-MJX (4,096 parallel environments, H100 GPUs) learned swing-up and balancing on cart pole, double/triple pendulum, and a planar humanoid with no extrinsic reward, where intrinsic-motivation baselines (DIAYN, SMM, ICM, APT) fail, and upright hopping/running on Hopper/Walker with only a velocity term.
- **Realizable information-bottleneck model reduction** (CDC 2026, first author): constrained the Past–Future Information Bottleneck (PF-IB) to the block-Hankel manifold, proved it forms a Lie group, and derived a Lie-algebraic optimizer whose every iterate is a valid state-space model; over 5× lower prediction error than standard PF-IB and near-optimal LQR cost from compressed models.
- **Empowerment-based MPC** (ACC 2027, under review; co-first author): contributed to a framework unifying empowerment estimation and control in one action distribution, enabling zero-order MPC (CEM, MPPI) with only first-order derivatives; achieved self-recovery of a fallen hopper with no extrinsic reward and raised sparse-reward swing-up success from 0% to up to 100%.

San Jose State University — **San Jose, California**
Graduate Research Assistant — *September 2023 – May 2024*
- Generalized CCA to Lie groups (**ICCA**; ACC 2024, first author) using intrinsic Riemannian means and one-parameter-subgroup geodesics, guaranteeing predictions stay on the manifold; on a 14-joint MuJoCo robotic hand (SO(3)/SO(2) joints, 15,000 samples) cut test reconstruction error by 23% vs. Euclidean CCA and the train–test gap from 16% to 4%. Code: github.com/JWK7/ICCA.

Instructional Student Assistant — *February 2022 – May 2022*
- Taught lab sessions, developed exercises, and graded for a course on reinforcement learning in engineering.

University of California, Davis — **Davis, California**
Research Assistant — *June 2021 – January 2022*

Technical Skills

Languages: Python, Julia, C/C++, R, SQL, LaTeX
ML & Simulation: PyTorch, JAX, Keras, NumPy/SciPy, MuJoCo & MuJoCo-MJX, DeepMind Control Suite, OpenAI Gym
Developer Tools: Git, Slurm (HPC clusters), CUDA, Linux, Anaconda, AWS EC2
Methods: RL (PPO, DDPG, intrinsic motivation), MPC (CEM, MPPI), LQR, information theory (information bottleneck, empowerment, Lyapunov exponents / KS entropy), Lie groups & Riemannian optimization, deep learning, generative modeling, computer vision, NLP

Certifications: DeepLearning.AI Generative Adversarial Networks (GANs) Specialization